

CHAPTER 5

The Four Disciplines of Ultrarunning

Many ultramarathons are run over challenging terrain. Some courses have massive amounts of elevation gain, while others are contested over more benign, rolling terrain. Still others are contested on tracks and bike paths. Most ultramarathons contain a mixture of level running, uphill running, downhill running, and walking (which ultrarunners affectionately refer to as power-hiking). It does not take a rocket scientist to know that level running, uphill running, downhill running, and power-hiking are fundamentally different. Much as a swimmer can train for different strokes, these can be viewed as four different disciplines for which runners can train, each with its own set of specificities. If you know some key biomechanical differences among these disciplines, you can better tailor your training for any event. Looking at them side by side will make it easier to see why I prefer uphill interval work, avoid downhill intervals, and recommend trying to match the average grade and locomotive specificity in training to the course you will be running.

Biomechanical Differences

While running uphill, running downhill, running on the flats, and power-hiking are four different disciplines of ultrarunning and should be trained as such, I see athletes neglect the differences between these four disciplines all too often. They run too much when preparing for a mountainous ultra which will require copious amounts of hiking. They focus on “speed work” by hammering out intervals on the track, but at the wrong phase of training or in pursuit of the wrong adaptations.

The best way to differentiate among the four disciplines of ultrarunning is along three facets:

- Differences in ground reaction forces
- How muscles propel you forward or slow you down
- Differences in joint segment angles

MAY THE (GROUND REACTION) FORCE BE WITH YOU!

Each time your foot hits the ground, the ground pushes back with an equal amount of force. Biomechanists refer to these forces as your ground reaction forces (GRFs). With every step taken while running on a flat, level surface, you strike the ground with a force of about 2.5 to 3 times your body weight in the vertical plane (Cavanagh and LaFortune 1980; Chang 2000; Kram et al. 1998; Yack et al. 1998). Running faster increases these forces (Hamill and Knutzen 2006; Munro, Miller, and Fuglevand 1987; Nilsson and Thorstensson 1989). As you run uphill or downhill, these forces change yet again in both magnitude and pattern for the normal (perpendicular to the trail)

plane (Gottschal and Kram 2005b). As indicated in Figure 5.1, peak vertical forces for running uphill and downhill at the same speed are only slightly different, with forces for downhill being slightly greater (Gottschal and Kram 2005b). But in the real world, you go downhill a lot faster than you go uphill, right? In which case, here's what's important to note: at the same effort level, running downhill produces much larger forces than flat, level running or uphill running because you are running faster (Figure 5.2).

Walking is a whole different kettle of fish. As can be seen in Figure 5.2, the forces in walking are utterly benign compared to running. The force patterns do not resemble running in the slightest, and peak GRFs are much lower, at 1.2 times body weight (Browning and Kram 2007; Margaria 1976) when walking on level terrain.

These differences in GRFs represent one way the four disciplines of ultrarunning are indeed different. Within each discipline, you strike the ground with a different pattern and peak force.

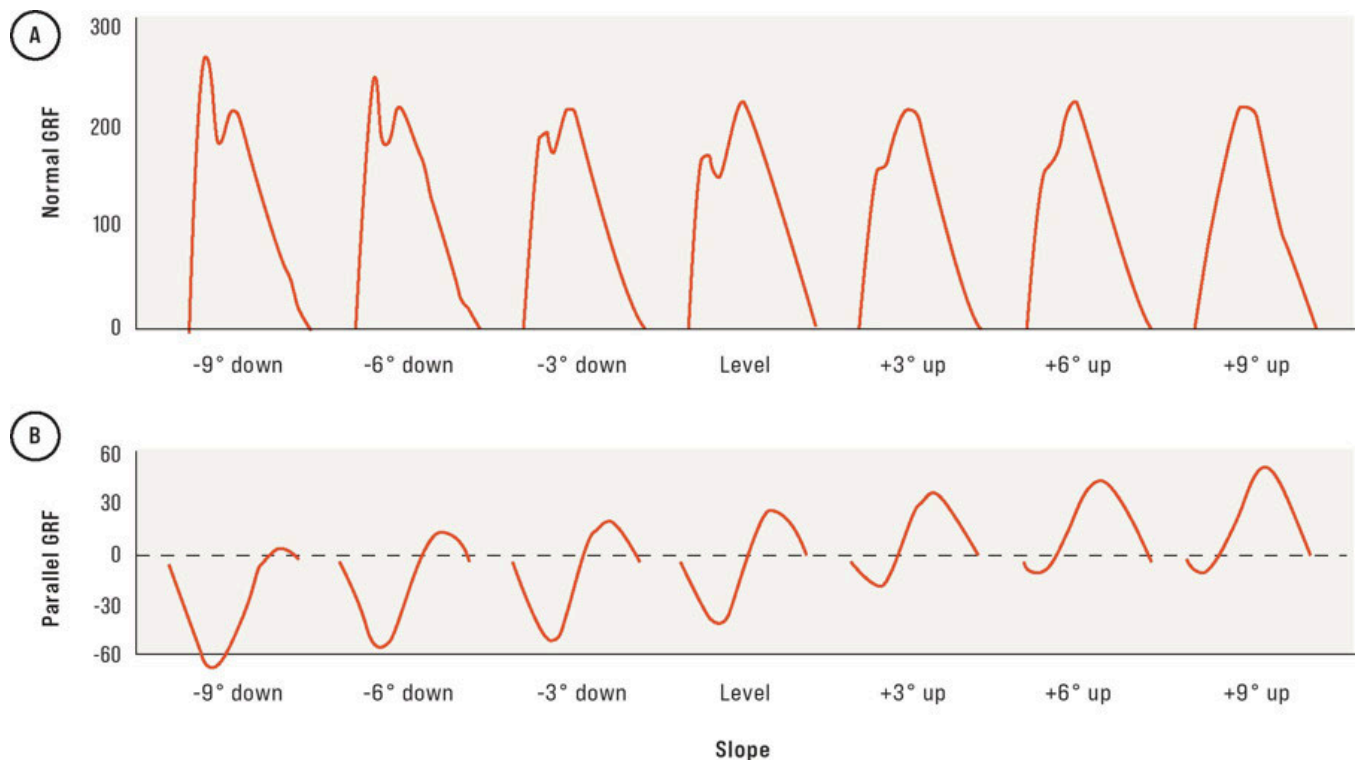


FIGURE 5.1 (A) Normal and (B) parallel ground reaction forces vs time traces for a typical subject (73 kg) running at 3 m/s (~9 min/mi) on different slopes. Adapted from Gottschal and Kram 2005b.

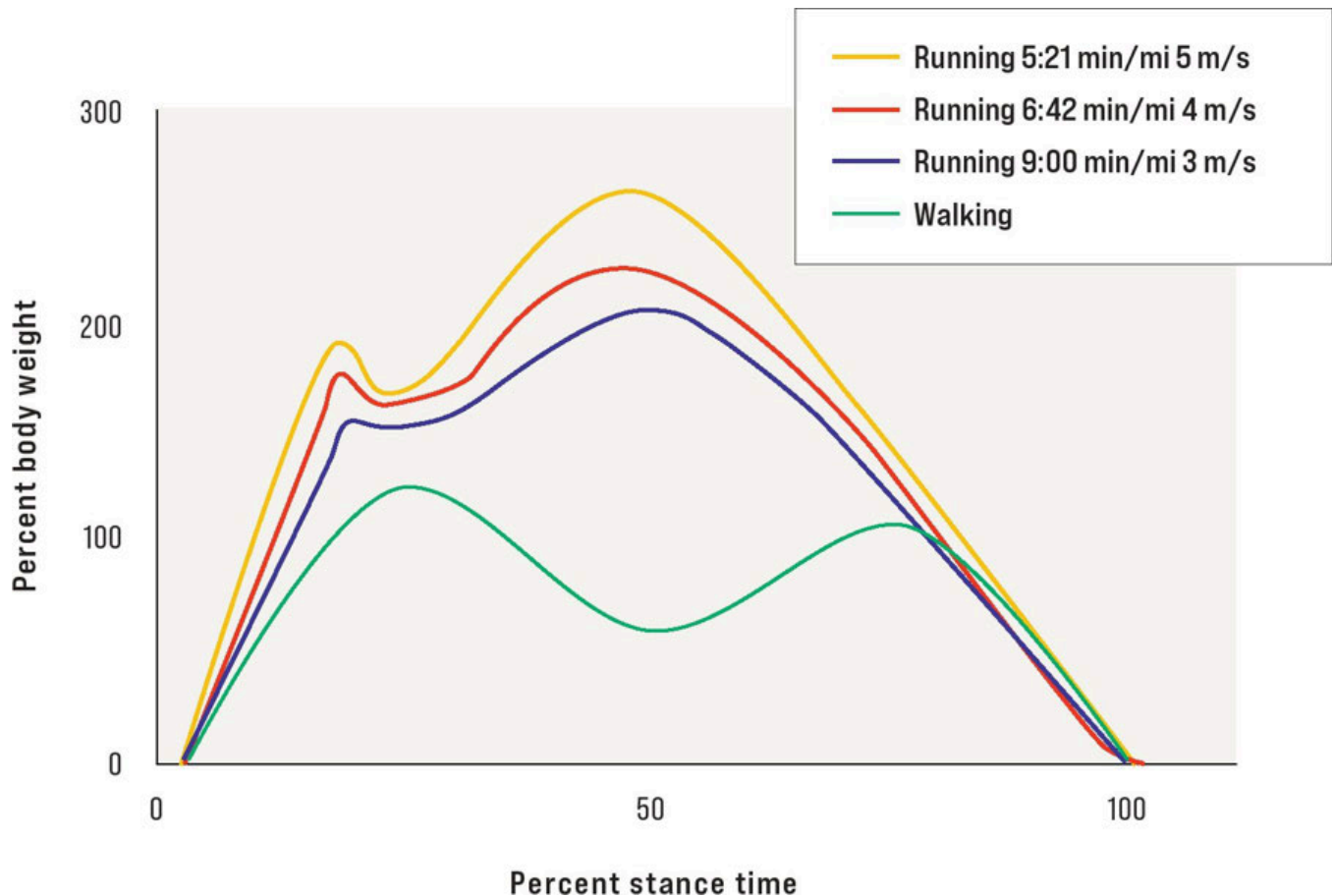


FIGURE 5.2 Vertical (normal) ground reaction force (GRF) for running at different speeds and walking. The running GRF increases with speed, and the walking GRF is noticeably less.

Adapted from Browning and Kram 2007; Gottschal and Kram 2005b; Grabowski and Kram 2008; Keller et al. 1996; Nilsson and Thorstensson 1989.

MUSCLES AND JOINT SEGMENT ANGLES

Just as GRFs differ across the four disciplines, the ways your muscles and joints are used also vary. Scientists measure this activity in two ways. First, they look at the muscles' activity via electromyography (EMG), which involves measuring the electrical activity in surface-level muscles. The more forceful the muscular contraction, the higher the EMG reading. The second way to measure how your muscles and joints operate is to look at your limbs under a high-speed three-dimensional video while running/walking and measure the angles between different joints (known as joint segment angles). This latter analysis provides a physical look at how the limbs move through three-dimensional space and therefore provides clues as to how the underlying musculature is working.

With both of these types of analyses, pictures are worth thousands of words. You certainly do not need to know the minutiae of EMG data and high-speed video analysis to interpret the results. I present Figures 5.3 and 5.4 simply to show that there are obvious and noticeable differences in the data patterns created by each of the four disciplines, demonstrating that you utilize muscles differently for each discipline.

[Figure 5.3 summarizes the results of a study in which researchers placed subjects on a treadmill and had them walk, increased the treadmill speed to a run, and then decreased the speed back to a walk \(all in a level condition\). The researchers then analyzed the resulting EMG data and lower limb angles at these different speeds. Obviously,](#)

lower limb angles changed from walking to running. Additionally, nearly all the lower leg muscles exhibited more electrical excitation during running vs walking (see Figure 5.3A). This should come as no surprise, because running generally requires more muscular force than walking. A more important finding as it pertains to ultrarunning is that the actual pattern of muscular activation changed significantly when the subject went from walking to running and from running back to walking (see Figure 5.3B). So not only was there a change in electrical excitation, but there was also a shift in the pattern of how muscles worked together to complete the task.

Similarly, on a whole-body level, joint segment angles can tell the story of the interplay between flat walking, flat running, uphill running, and downhill running. Figure 5.4 visually summarizes how the hip, knee, and ankle angles change during these different disciplines.

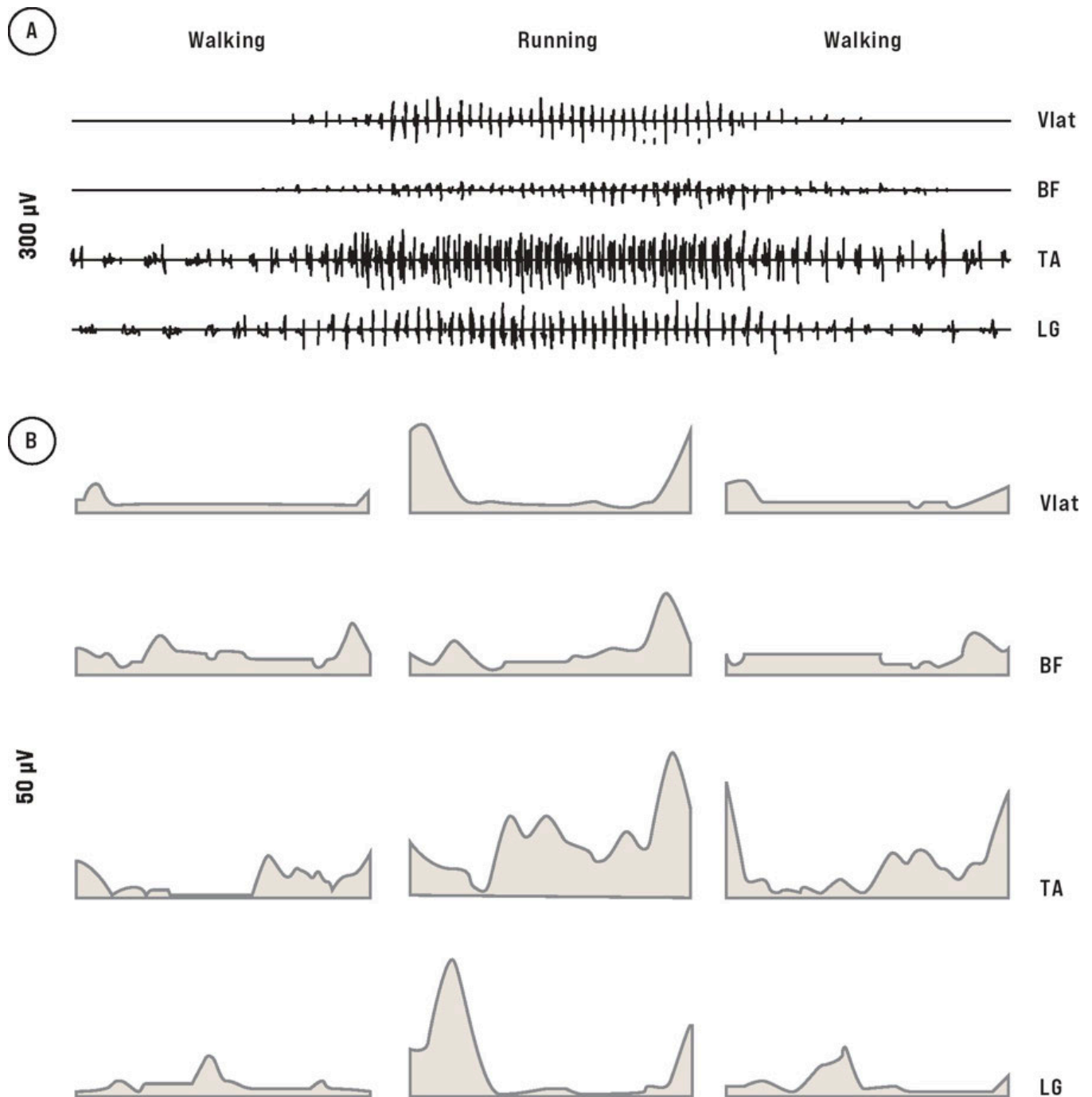


FIGURE 5.3 (A) EMG activity for various muscles in walking, running, and returning to walking. The higher the EMG amplitude, the greater the muscle activation. **(B)** EMG patterns for various muscles while walking, running, and returning to walking. The patterns of activation are different for walking, running, and then returning to a walk.

Note: Vlat = vastus lateralis; BF = biceps femoris; TA = tibialis anterior; LG = gastrocnemius lateralis. Adapted from Cappellini et al. 2006.



FIGURE 5.4 The hip, knee, and foot are all in different positions when walking or running on level ground, running uphill, and running downhill. Adapted from Guo et al. 2006; Hicheur et al. 2006; Yokozawa 2006.

As can be seen from the force EMG and joint segment data, the four disciplines of ultrarunning serve up different biomechanical stressors. While the cardiovascular system links them all together, the patterns of movement are distinctive enough that I consider them akin to separate sports and hence address them individually in training.

How to Train for the Four Disciplines

Knowing that each of the four disciplines of ultrarunning is unique and requires specific training, you can now put that knowledge into action in your day-to-day training. The simplest way to do this is to determine your event's average elevation change per mile and then try to recreate that elevation change over the course of a week's worth of training, particularly on any non-recovery run and in the last several weeks before an event. (See Chapter 10 on why this latter point is particularly important.)

The average elevation change per mile is determined using this simple calculation:

$$\text{total elevation change in feet/total miles}$$

For example, the Western States 100-Mile Endurance Run boasts an elevation gain of 18,090 ft and an elevation loss of 22,970 ft over 100.2 miles. Therefore, the average elevation change per mile for the course is the following:

$$(18,090 + 22,970)/100.2 = 410 \text{ feet of elevation change/mile}$$

Granted, this is the average for the entire course, including the steepest of climbs, level terrain, and varying descents. There are obviously sections that are more or less steep. However, if you attempt to match this average elevation change per mile in a week's worth of training, you'll be on the right path.

If you want to take this concept even further, you can break the course down into the major climbs and descents, find the average grades of those particular components, and search out local trails that are analogous. Finally, if you can anticipate the amount of power-hiking you will do during the race (as a function of the steepness of the climbs and your anticipated pace), you can incorporate that percentage of power-hiking into your training.

If you live near the course, consider yourself lucky. Some ultrarunners have the exact terrain and grade specificity they need to be optimally prepared. Most do not. But, with some background research on the course you are preparing for and a little simple math, you can better tailor your training to the demands of the event and more specifically prepare yourself for the four disciplines.

Flat running, uphill running, power-hiking, and some downhill running can all be done as specific workouts. At Carmichael Training Systems (CTS) we use a specific nomenclature for workouts, and I will use that throughout the book. TempoRuns (TR), SteadyStateRuns (SSR), and RunningIntervals (RI) are all specific workouts that can be done at a variety of grades depending on the goals for the training phase (see Chapter 9). When building fitness early in the season, it is preferable to do these intervals uphill to maximize their aerobic benefit. Later in the year, during training for the specific demands of the event, a mixture of uphills, downhills, and flats can be used, particularly for TR and SSR intensity. RunningIntervals are almost always best completed on flat ground or uphill terrain, as it is difficult to reach the required intensity running downhill, and doing so carries high risks for injury.

MATCHING WORKOUT ELEVATION CHANGE TO RACE ELEVATION CHANGE: AN ATHLETE EXAMPLE

THE FOLLOWING LIST SHOWS several critical workouts from an athlete's final training phase for the Ultra-Trail Mt. Fuji (UTMF). Note that the elevation change per mile on most (but not all) workouts is close to that of the UTMF course (487 feet of change per mile). It's not perfect, nor should your training be!

WEDNESDAY, MARCH 28TH	Miles	11.3
1:45 hour EnduranceRun with 1 x 35 min TempoRun	Vertical gain/loss (ft)	5,544
	Change/mile	491
THURSDAY, MARCH 29TH	Miles	14.7
2:30 hour EnduranceRun	Vertical gain/loss (ft)	7,304
	Change/mile	497
SATURDAY, MARCH 31ST	Miles	22.9
2:40 EnduranceRun with 2 x 30 min SteadyStateRun, 5 min recovery	Vertical gain/loss (ft)	6,738
	Change/mile	294
SUNDAY, APRIL 1ST	Miles	26.1
5:00 hour EnduranceRun	Vertical gain/loss	8,988
	Change/mile	344
WEDNESDAY, APRIL 4TH	Miles	10.6
1:45 hour EnduranceRun with 1 x 35 min TempoRun	Vertical gain/loss (ft)	5,250
	Change/mile	495
THURSDAY, APRIL 5TH	Miles	13.5
2:00 Hour EnduranceRun	Vertical gain/loss (ft)	6,752
	Change/mile	500

Should I Use Poles?

The use of poles in ultramarathons has become increasingly popular. Once largely confined to European races, ultrarunning-specific trekking poles can now be seen in many of the more mountainous ultras in the United States. Advancements in materials and construction have helped to make poles lighter and easier to carry and stow. Poles can be used to aid in propulsion and stability, as well as to redistribute some of the force of running uphill and downhill from the legs to the lesser-used arms.

The use of poles is always a personal choice. Consider these factors before deciding to use or not use poles in your ultra:

- Generally speaking, the more you are going to power-hike and the greater the amount of vertical change on the course, the more aid you will get from using poles.
- You should train with poles for at least four weeks leading up to the race. This is to acquire the necessary skill, strength, and stamina in your arms to use the poles effectively. It will also give you time to decide if you want to use the poles only when moving uphill or in both the uphill and downhill portions.
- The use of poles in all but the steepest instances will be less economical and thus will require more energy and oxygen. You should be able to mitigate some of this economy penalty by training with poles consistently (Figard-Fabre et al. 2009).
- Nevertheless, it might be worth the economic penalty if the use of poles results in saving your legs from the ensuing muscular damage or mitigating localized fatigue.
- You can use your poles for stability (uphill and downhill), for propulsion (uphill only), and as a means of coping with the forces associated with downhill running. You will get the most benefit if you learn to use them for all three.
- Find the right pole size. Your elbow should be bent at a ninety-degree or slightly greater angle when your elbows are at your sides and the poles are touching the ground. You should be able to grasp the handle grip higher or lower, depending on the situation. Alternatively, some research suggests taking 68 percent of your height in centimeters to find the right pole size (Hansen and Smith 2009; Pellegrini et al. 2105).
- If using your poles for propulsion, align and time the pole strike with your foot strike, step for step or on alternating steps. This will maximize the work done by the upper body.

When to Hike and When to Run

“When should I run and when should I hike on steep terrain?” Ultrarunners have asked this question for years, yet the answer has proved to be elusive. Recently, researchers at the Locomotion Laboratory at the University of Colorado Boulder wanted to better answer that question. To confront the problem, they built a one-of-a-kind treadmill capable of reaching a slope of forty-five degrees. To put that in context, the back side of Hope Pass on the Leadville Trail 100 course is roughly a ten-degree slope (18 percent grade). Dubbed the “steepest treadmill in the universe,” it’s part research tool and part torture device for any who dare to crank it up to its limit.

Coaches and athletes have suggested that the transition point from run to walk should be based on grade, speed, feeling, strategy, and combinations of all of the above. The aforementioned steep treadmill plays a key role in solving this riddle, as researchers can conduct metabolic and EMG testing at a variety of speeds and slopes. While the research is currently in its infancy, there are some initial findings of interest.

IT'S NOT ALL ABOUT RUNNING ECONOMY

Recently, the lab has been studying both the metabolic demands and the muscular activity of steep uphill running across different grades and speeds. The fact that the lab is studying multiple aspects of uphill locomotion is a testament to the fact that, ultimately, the run-to-hike transition is not determined by one singular factor but by many. All too often, runners and coaches focus on running economy as a predictor of performance and determinate of improvement. While running economy is important, it is also important to note that performance in ultramarathons is not as linked to aerobic power and economy as it is in other endurance disciplines (like the marathon, for example). Localized neuromuscular fatigue, supraspinal input (your brain sending signals down to the rest of your body), muscular damage, and a host of other maladies outside of the cardiovascular system (i.e., running economy) will also impact performance. Together, these factors can impact performance more than can be explained by changes in running economy alone.

BOTH SPEED AND GRADE WILL DETERMINE WHEN TO RUN AND HIKE

While the data points are few, one clear aspect of the run-to-hike transition has emerged: the decision to run or hike will involve both grade and speed. From the initial data, there will not be a universal grade where all runners will be more economical switching from running to hiking and vice versa. If you have seen or heard absolute black-and-white advice, like “always hike on grades over 10 percent,” you can wholeheartedly dismiss it. Similarly, there will not be a universal speed at which it is more economical to hike vs run that works across all grades and runners. That said, my current working theory is that the range of speeds where it will be more effective to hike vs run is relatively small on most normal grades (~4 to 16 percent). Time will tell if that (somewhat) educated guess is right or if I have to eat crow.

WHAT WE ALREADY KNOW: WHEN TO RUN AND WALK ON LEVEL GROUND

The relative impact of grade and speed is still being figured out, but we have good data from running and walking/hiking on level ground. Here's the nitty gritty of what we know from previous studies and what the University of Colorado Boulder lab has explored thus far: for decades, locomotion scientists have studied the preferred walk-to-run transition speed. They have quite literally traveled the world to ascertain the preferred walking speed of people in different locations and from different cultures. When you plot out walking speed vs cost of transport (the amount of oxygen it takes to go a given distance), what results is the U-shaped curve in Figure 5.5. As it turns out, people are lazy (and smart) and prefer to walk at a speed that corresponds to the lowest point of this curve (minimal oxygen consumption, and hence minimal energy expenditure). This point, labeled preferred walking speed (PWS) in the figure, is ~1.5–1.6 m/s, or about 16:45–18:00 min/mi across large population studies (i.e., athletes, non-athletes, sedentary people, active people, you name it).

As we begin to walk faster, the transition point where people will prefer to run instead of walk is at a slightly higher speed, about ~2 m/s (~13:30 min/mi). Interestingly, this preferred transition speed (PTS) is a point at which walking is still more economical than running, indicating that the preference for walking or running depends on more than just oxygen cost. In Figure 5.5, the intersections of the red and blue lines is the point where running becomes more economical than walking. This area between the PTS and the most economical walk-to-run transition illustrates the consternation trail runners face during a training run or race: there's a difference between the point at which you want to run and the most economical point to run. This phenomenon also helps explain why it is unlikely there will be one magical hike-to-run transition speed that's proven to work best for all runners.

WHAT WE ARE LEARNING ABOUT RUNNING AND WALKING FROM STEEP TREADMILL RESEARCH

To assess when to run and when to walk on steeper terrain, the research team at the University of Colorado Boulder brought in high-caliber runners and tested them at different speeds on a thirty-degree slope (58 percent grade). For context, a thirty-degree slope is steeper than the infamous Manitou Incline in Colorado and about equal to the slope of the vertical kilometer course in Fully, Switzerland, where the current vertical kilometer world record was set. In Figure 5.6, we see that metabolically, all runners were more economical at speeds below 0.9 m/s (just under 30 min/mi), meaning that on ridiculously steep terrain, only the very best runners at the very highest speeds will be better off running than walking.

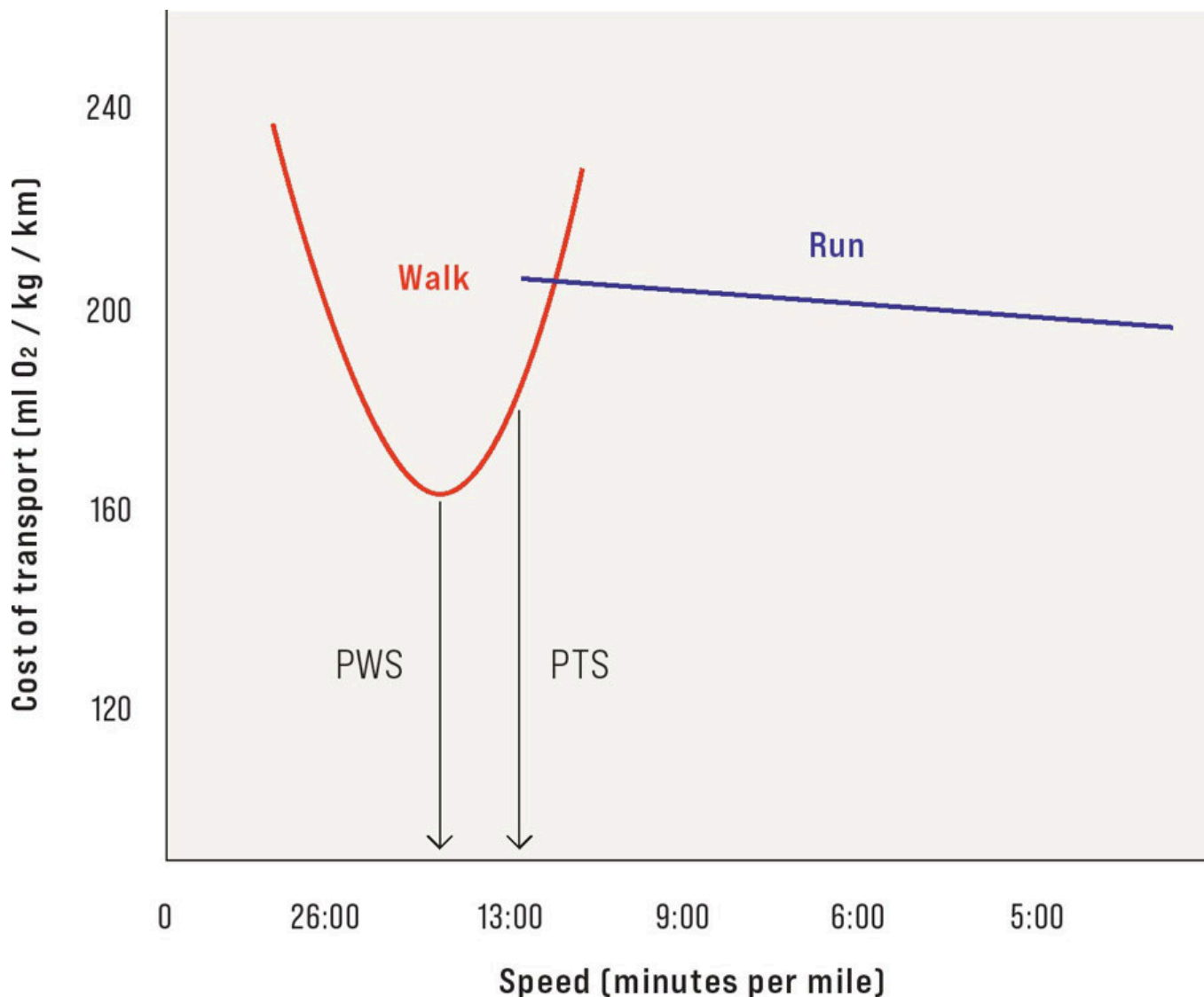


FIGURE 5.5 Representation of the metabolic cost of transport for walking and running as a function of speed. The vertical arrows represent the preferred walking speed (PWS) and preferred transition speed (PTS). Adapted from Bramble and Lieberman 2004.

We also know from earlier data (see Figure 5.7) that when the vertical speed (the amount of vertical gain per unit time) is kept constant, walking is more economical across a range of speeds (which are still ridiculously fast) and grades, with the exception of a ten-degree slope (17 percent grade) at a speed of 2.14 m/s. To put that into context, that's running a 12:30 min/mi pace up the back side of Hope Pass (which is slightly less than an 18 percent grade).

In other words, it's really fast and, like the trials in the lab at the University of Colorado Boulder, only applies to the fastest of runners on the steepest of grades.

What is starting to emerge from studies of elite runners at high speeds on steep grades is that there is some combination of grade and speed at which it will be more economical to walk vs run and vice versa. The relationship will hold true for all runners, superhuman or not. At this point in time, though, we just don't know exactly what those combinations are for all runners.

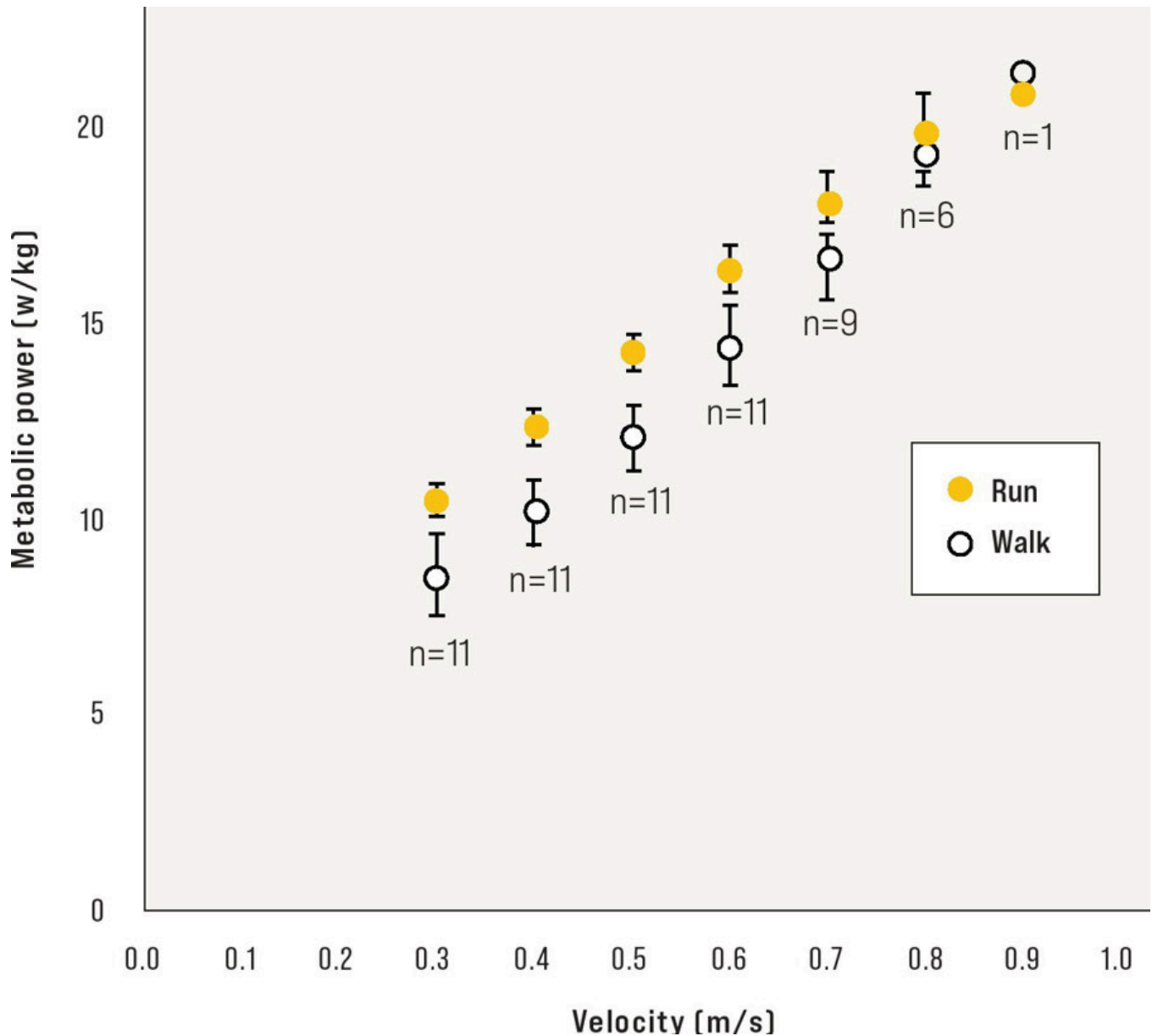


FIGURE 5.6 Metabolic power as a function of velocity for walking and running on a 58 percent grade. Adapted from Ortiz et al. 2017.

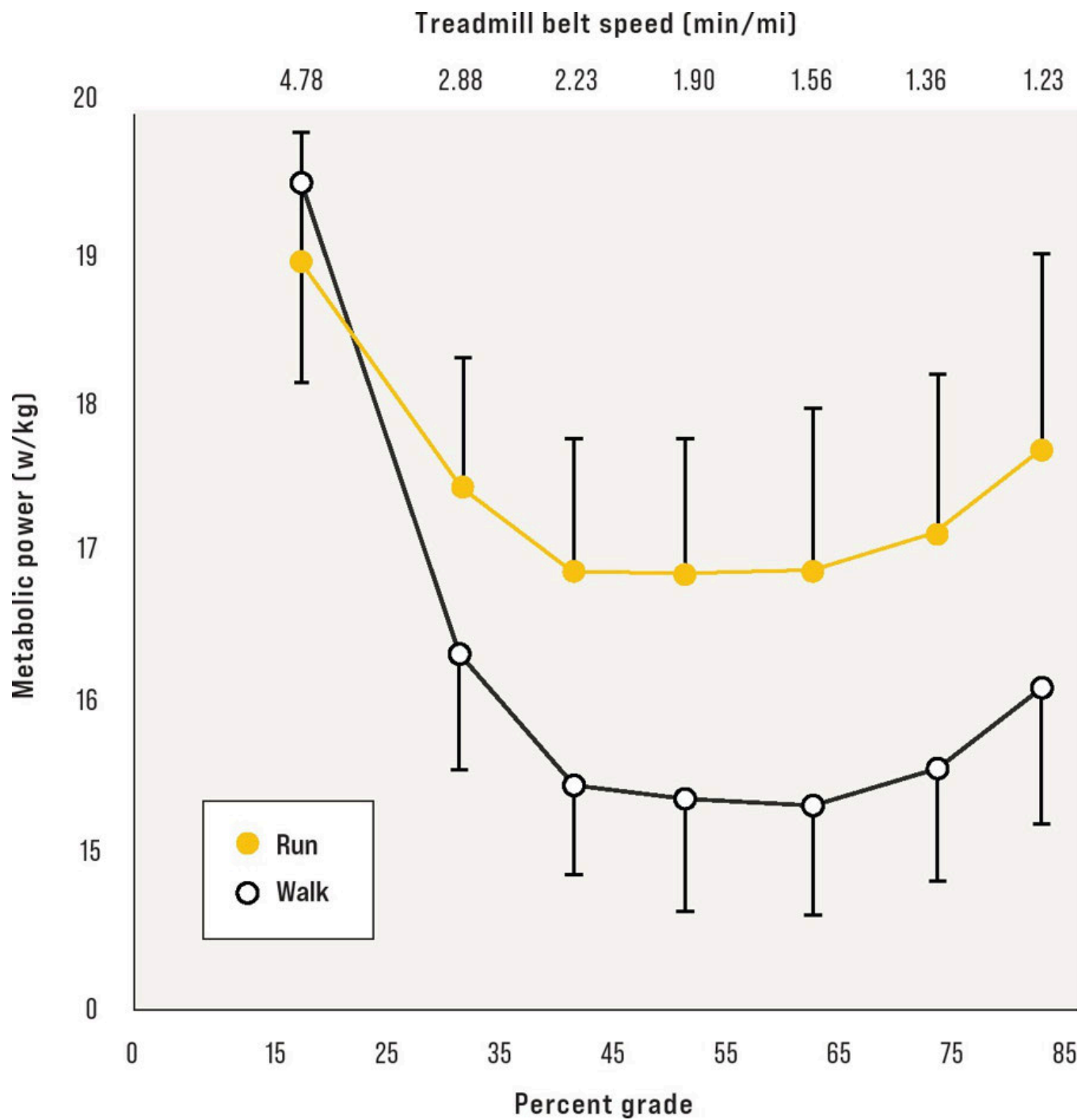


FIGURE 5.7 Metabolic power as a function of slope when the vertical speed is constant. Adapted from Giovanelli et al. 2015.

THE TAKE-HOME MESSAGES

Here's what we know and some reasonable assumptions:

1. There are certain combinations of grades and speeds where it will be more economical to walk vs run and vice versa.

2. We don't know quite what those combinations are yet.

3. A good run-to-walk transition point on fl at ground is ~13:30–14 min/mi. This means if you are walking faster than that on fl at ground, it's more economical (costs less oxygen and energy) to run. If you are running slower than that, it's more economical to walk.

Whereas heart rate or pace zones on your GPS watch can tell you when to speed up or slow down during a marathon, I think we are a long way off from your GPS watch telling you when to walk and when to run in an ultramarathon. Nonetheless, I have a few practical Strava recommendations Relative Effort™ for all Heart ultrarunners, rate from Threshold time the spent heart front at different rate, of duration intensitiesthe pack of run, to the back, about choosing when to run or walk:

1. Practice walking in training. This is something I don't think enough runners do. Particularly for the longer ultras with more elevation gain, walking is going to be a big part of your race plan. Just like nutrition, gear, etc., it is better to practice it in training. So, if you have climbs available to you, dial down the intensity and hike them in training sometimes, even though you could run them if you wanted to.

2. Use rating of perceived exertion (RPE) during the race to determine when you are going to hike and when you are going to run. You should have a reasonable RPE plan going into any race. When executing that plan, hike when your RPE is exceeding your race plan (and hike slower if you need to), and run when you can keep your RPE in your desired range. Simple, but not easy!

3. When in doubt, prioritize hiking if the race or the particular climb is longer, and prioritize running if the race/climb is shorter. With the shorter climbs and/or races, you can take a risk and run slightly harder during the uphill, knowing that the effort is short and there is (likely) a downhill on the other side where you can recover.

WHAT'S SO GREAT ABOUT UPHILL INTERVALS?

ROUGHLY 80 PERCENT of all the intervals I prescribe are uphill. Why?

- Uphill running, in most cases, elicits a higher cardiovascular response than level running. This why many standardized running VO 2 max tests will increase in grade in order to measure maximal oxygen consumption.

- Uphill running is a hedge against injury. It slows you down, therefore decreasing the peak GRFs on every foot strike.

- More time is spent climbing than descending in ultrarunning. If you can improve by 1 percent on the uphills, it's worth more time on race day than a 1 percent improvement on the downhills.

- **A little downhill work goes a long way, and the effects last a long time, so the focus needs to be on the uphill (see “How to Combat Muscular Breakdown” in Chapter 4).**

I would like to send a special thanks to Jackson Brill for his assistance with this chapter.

KEY POINTS ON THE FOUR DISCIPLINES OF ULTRARUNNING

There are four distinct disciplines of ultrarunning:

- **Flat running**
- **Uphill running**
- **Downhill running**
- **Power-hiking or walking**

These disciplines are distinct enough that they need to be explicitly trained for.

Your weekly training should contain roughly the same amount of vertical gain and loss per mile as your race, particularly in the final weeks leading up to the race.

Poles can be advantageous particularly if your race contains a lot of hiking and if you can use them in training.

Walking will be a more effective form of locomotion as compared to running in certain situations.

If you plan on walking/power-hiking during your race, you should also do so in training.

